

# Intro to Phylogenetics Using Nextstrain

Day 4 of Training Workshop

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# Section 1: Overview

# Getting Started

Before we get started, let's confirm your computer has **nextstrain** setup properly. Open a terminal and run the following command:

```
1 nextstrain setup --set-default conda
```



You should see the following output:

```
Checking setup...
```

- ✓ yes: operating system is supported
- ✓ yes: runtime data dir doesn't have spaces
- ✓ yes: runtime appears set up
- ✓ yes: snakemake is installed and runnable
- ✓ yes: augur is installed and runnable
- ✓ yes: auspice is installed and runnable

```
Setting default runtime to conda.
```

```
All good! Set up of conda complete.
```

# Getting Started (Cont.)

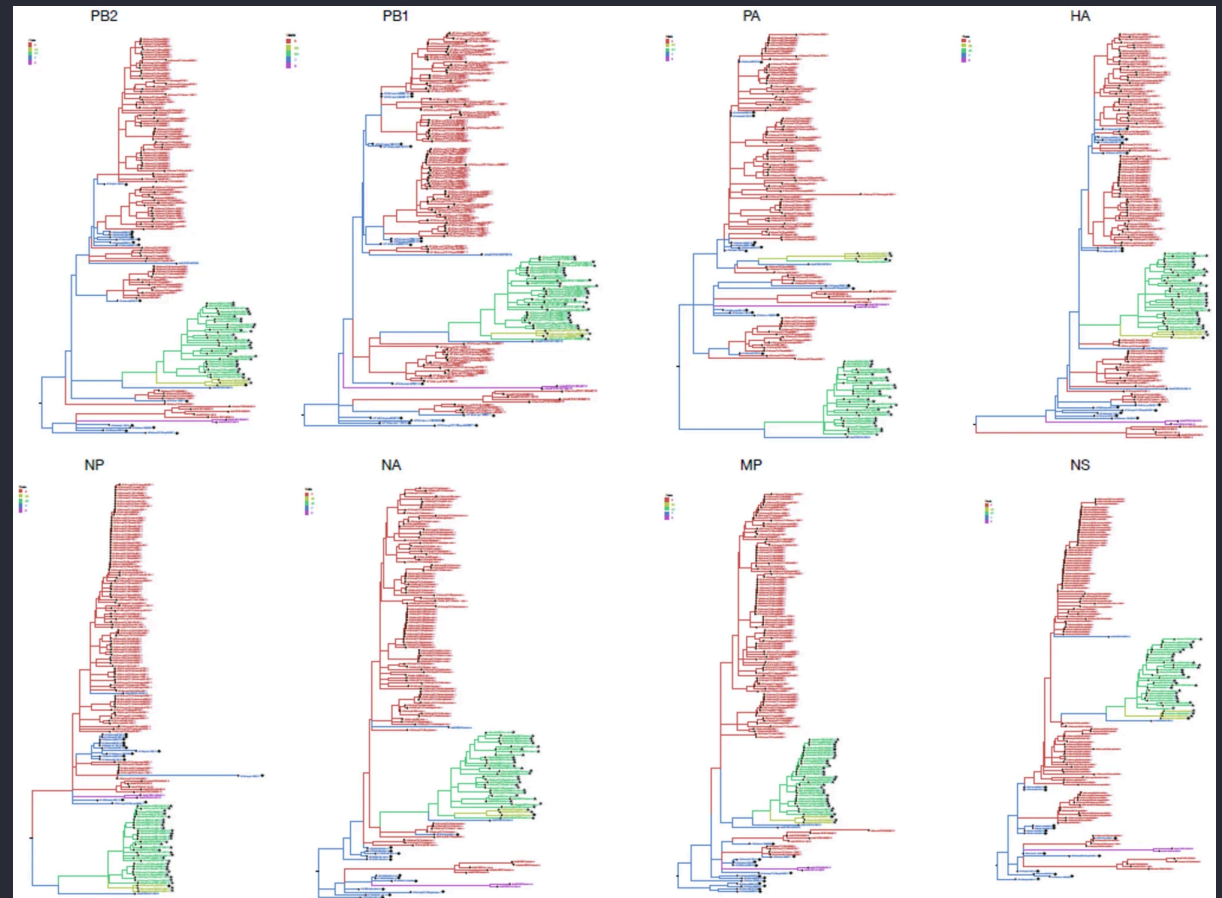
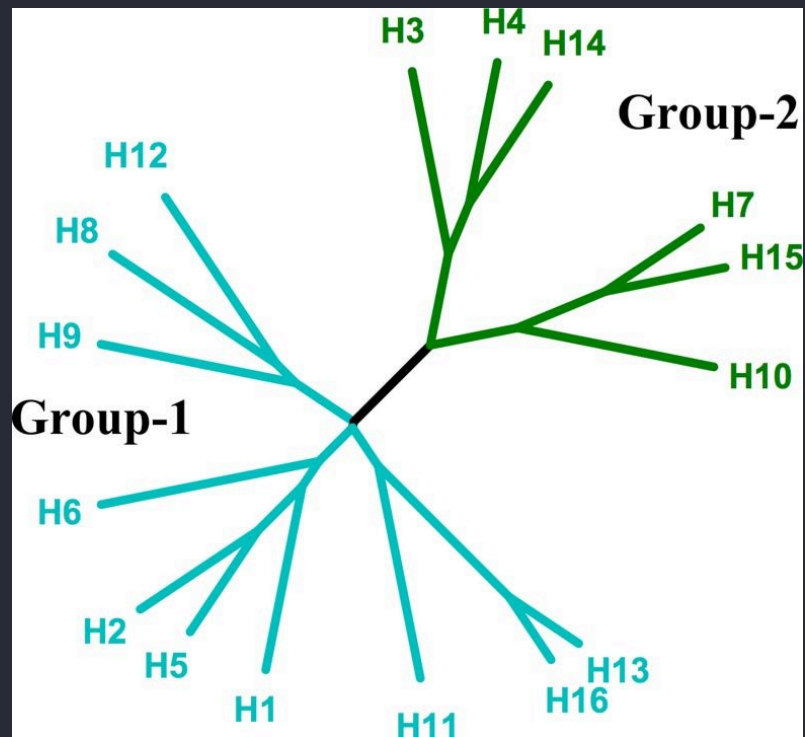
If you ran into an error when running the command, go to the following [website](#).

Install the `nextstrain-cli` using the instructions for your operating system.

For the “Set up a Nextstrain runtime” section, follow the “Conda” instructions for your operating system. We’ll go around to help if you are running into issues.

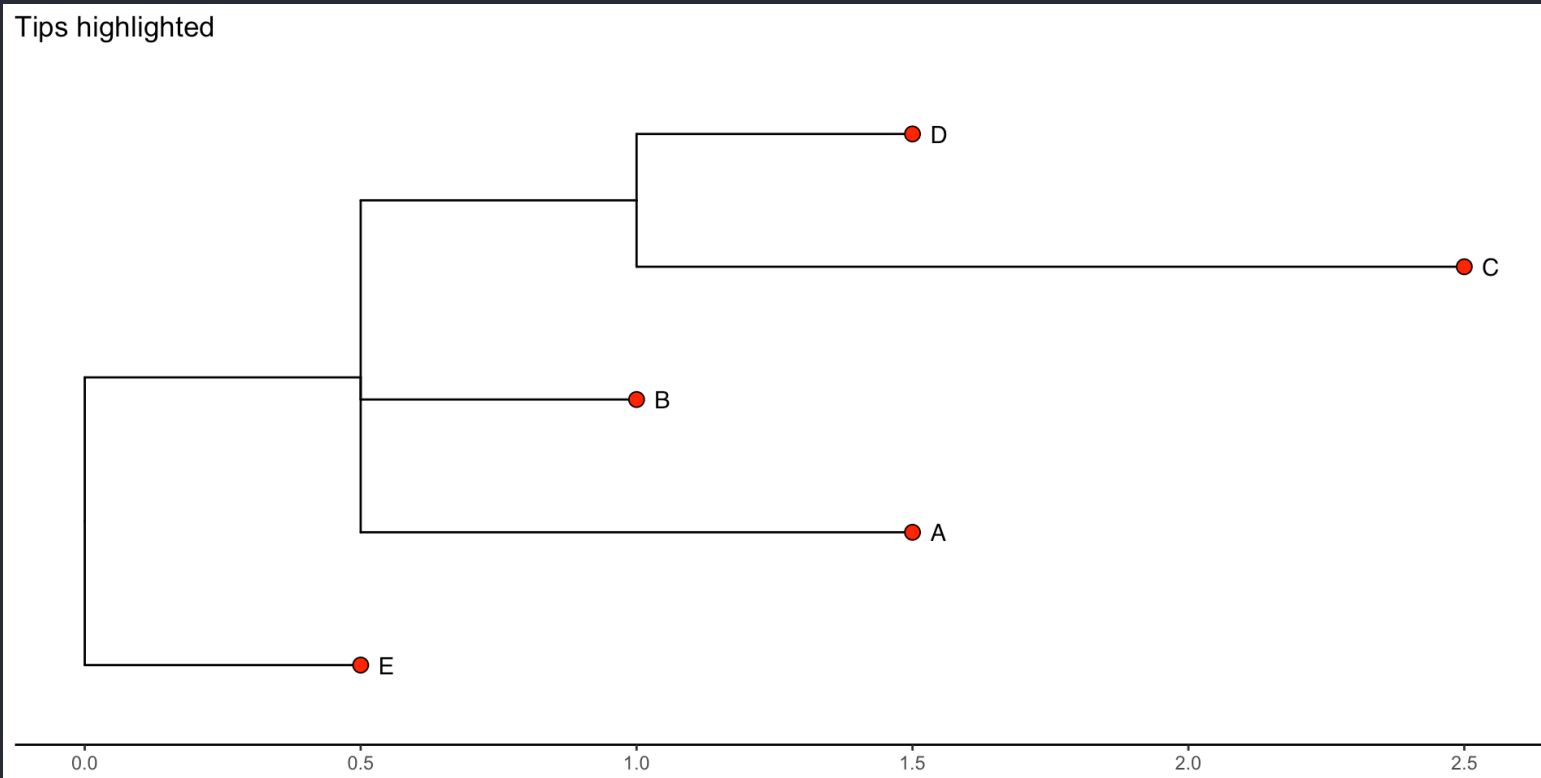
# Intro to the Intro

**Phylogenetics** is the study of evolutionary relationships among organisms.



# Basic Components of a Phylogenetic Tree (Tips)

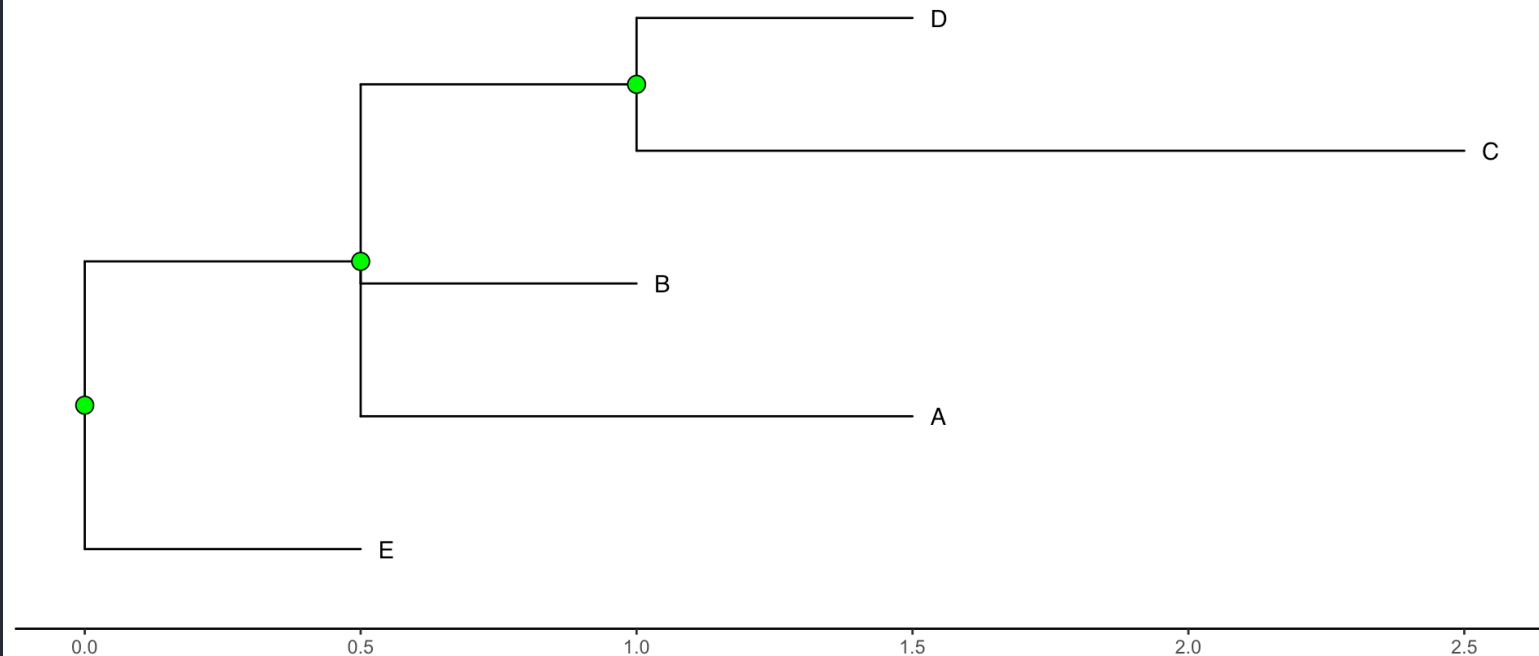
## ► Code



# Basic Components of a Phylogenetic Tree (Nodes)

## ► Code

Nodes highlighted

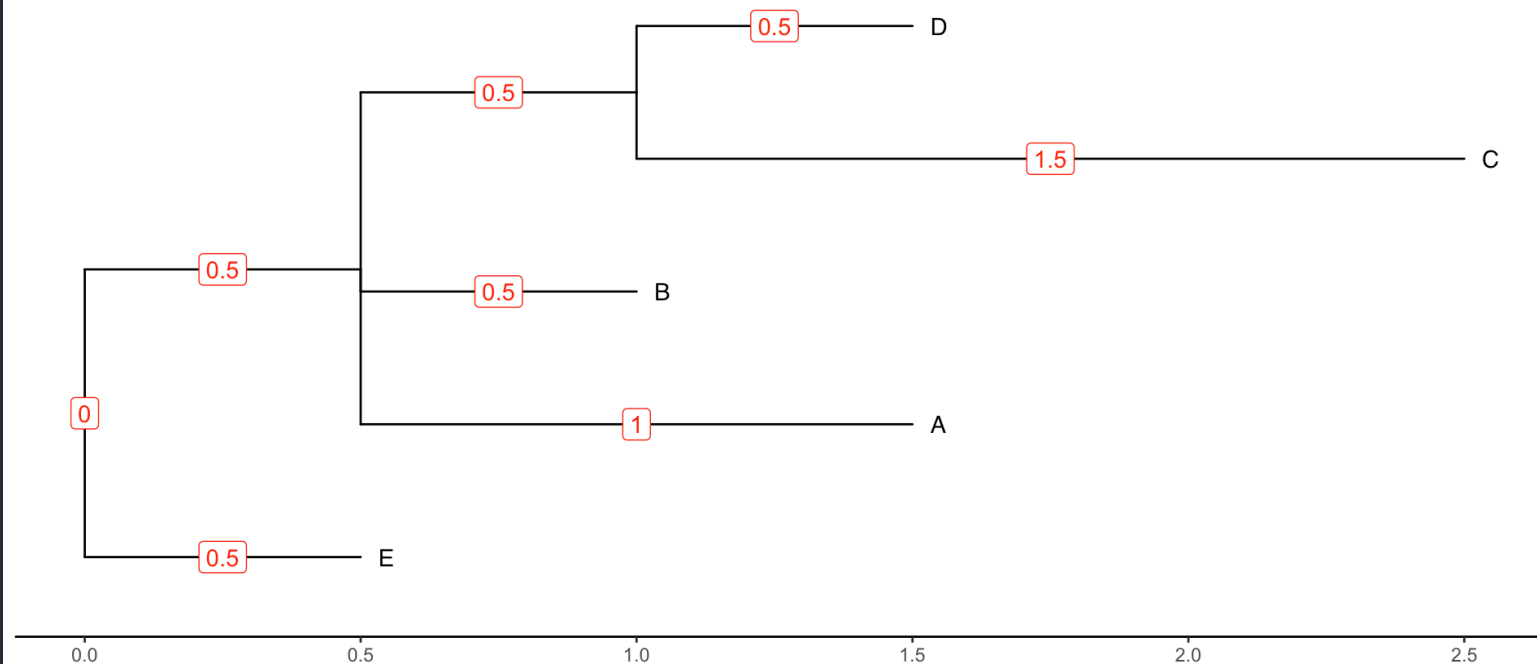




# Basic Components of a Phylogenetic Tree (Branches)

## ► Code

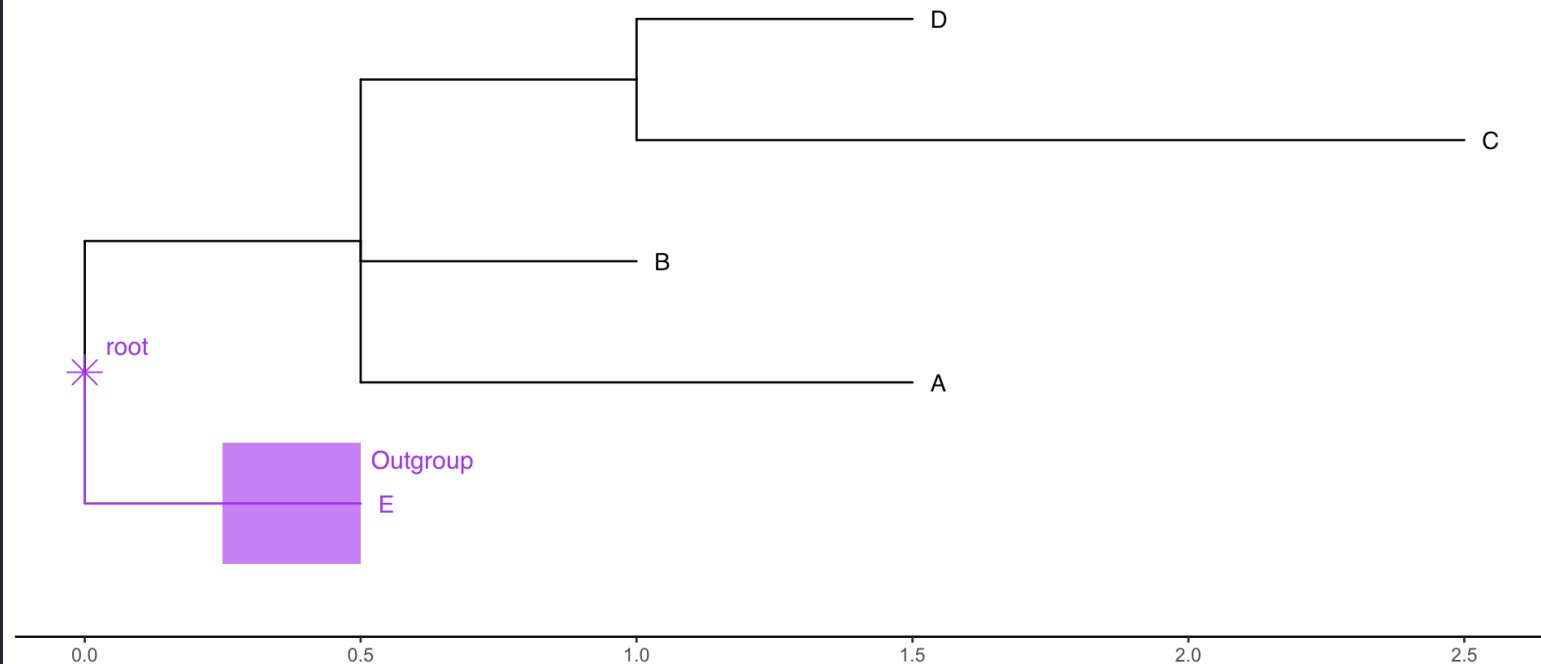
Branch lengths labeled (highlighted)



# Basic Components of a Phylogenetic Tree (Root)

## ► Code

Root and outgroup highlighted



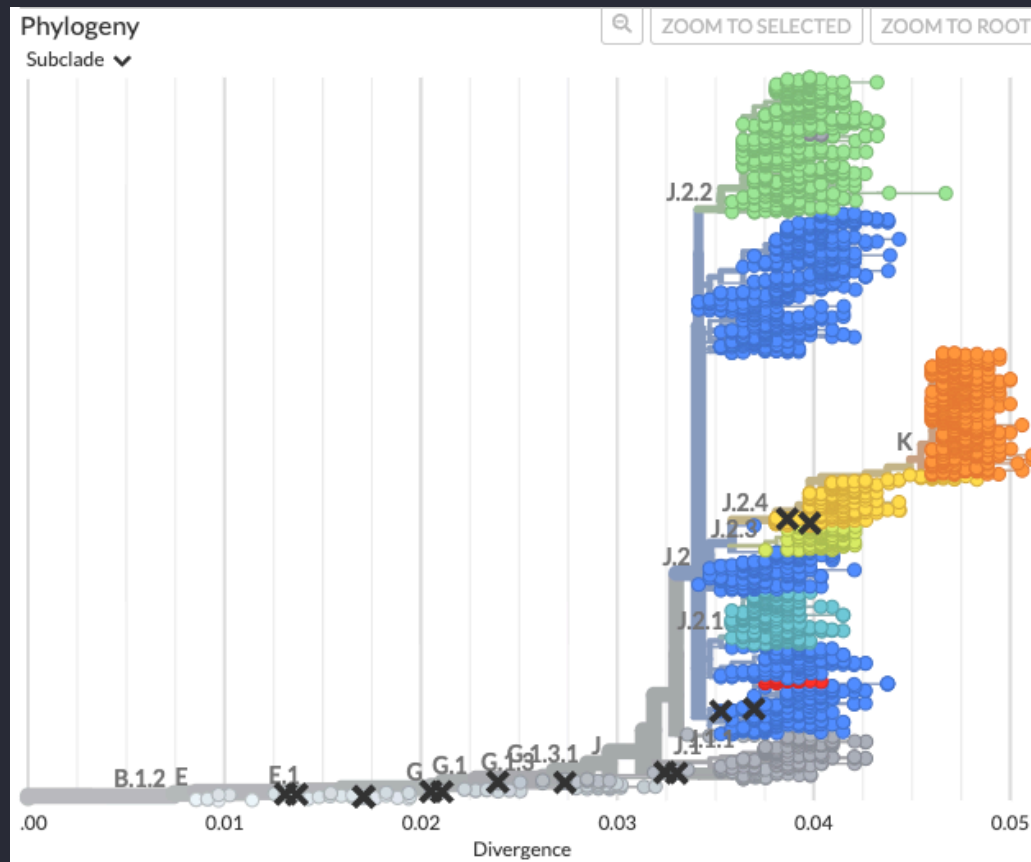
# Notes on Tree Rooting

Conceptually, tree rooting should be done:

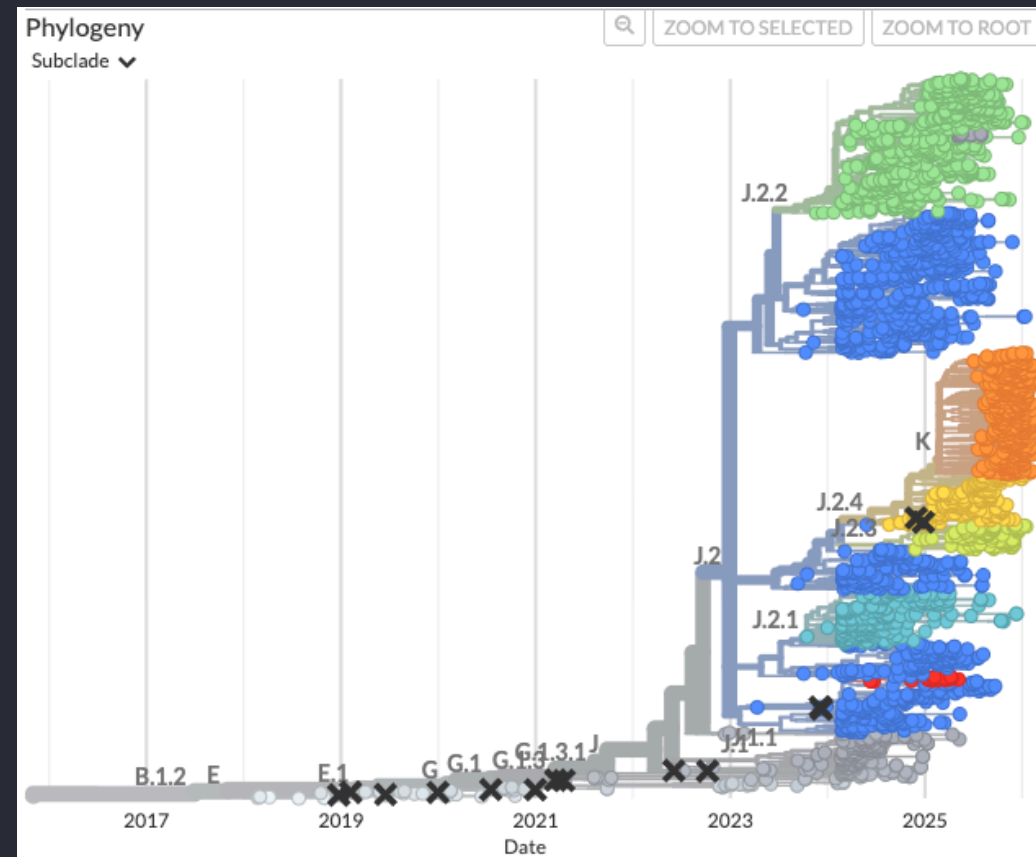
- Using a known outgroup (a taxon closely related to, but outside, the study group) to place the ancestor
- Or using midpoint rooting, which sets the root at the midpoint of the longest path between any two taxa

Outgroup rooting provides a more robust evolutionary history, and is generally preferred.

# Types of Trees

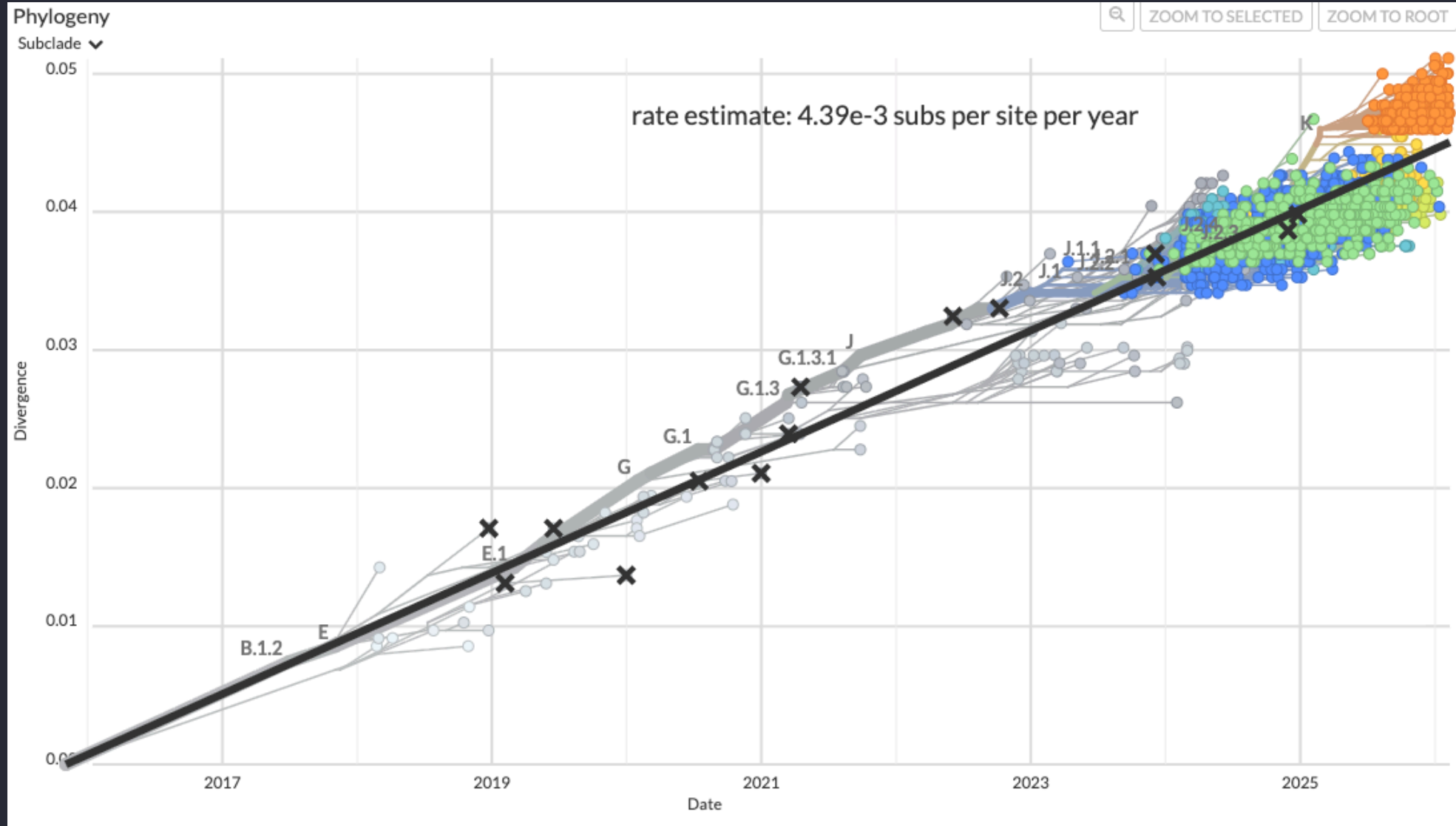


Nextstrain Tree in Divergence Display



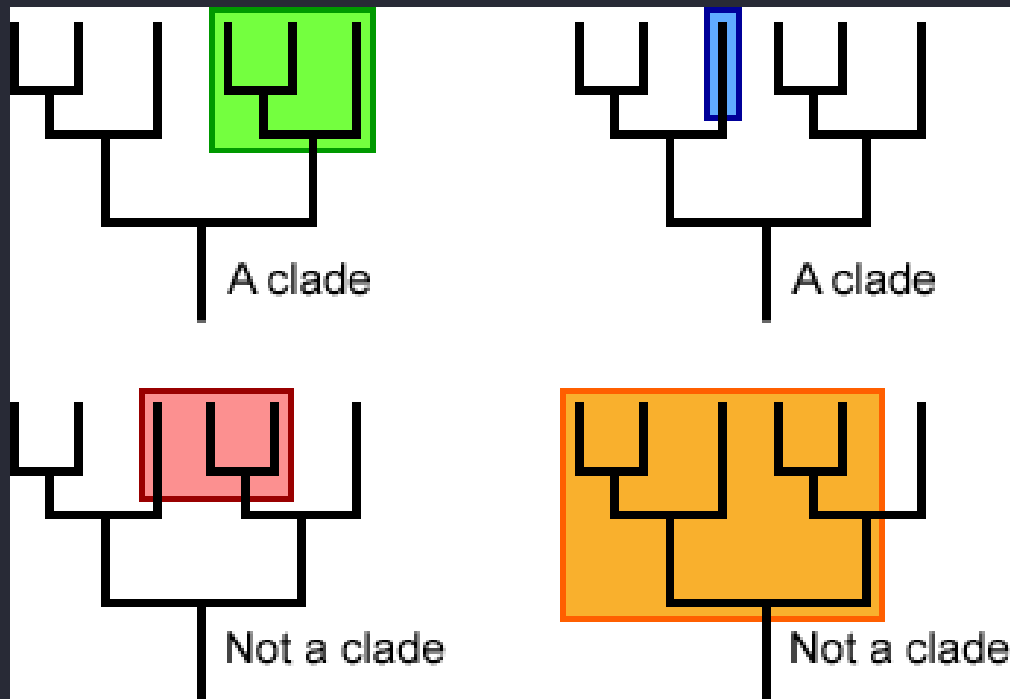
Nextstrain Tree in Time Display

# Molecular Clock Concept



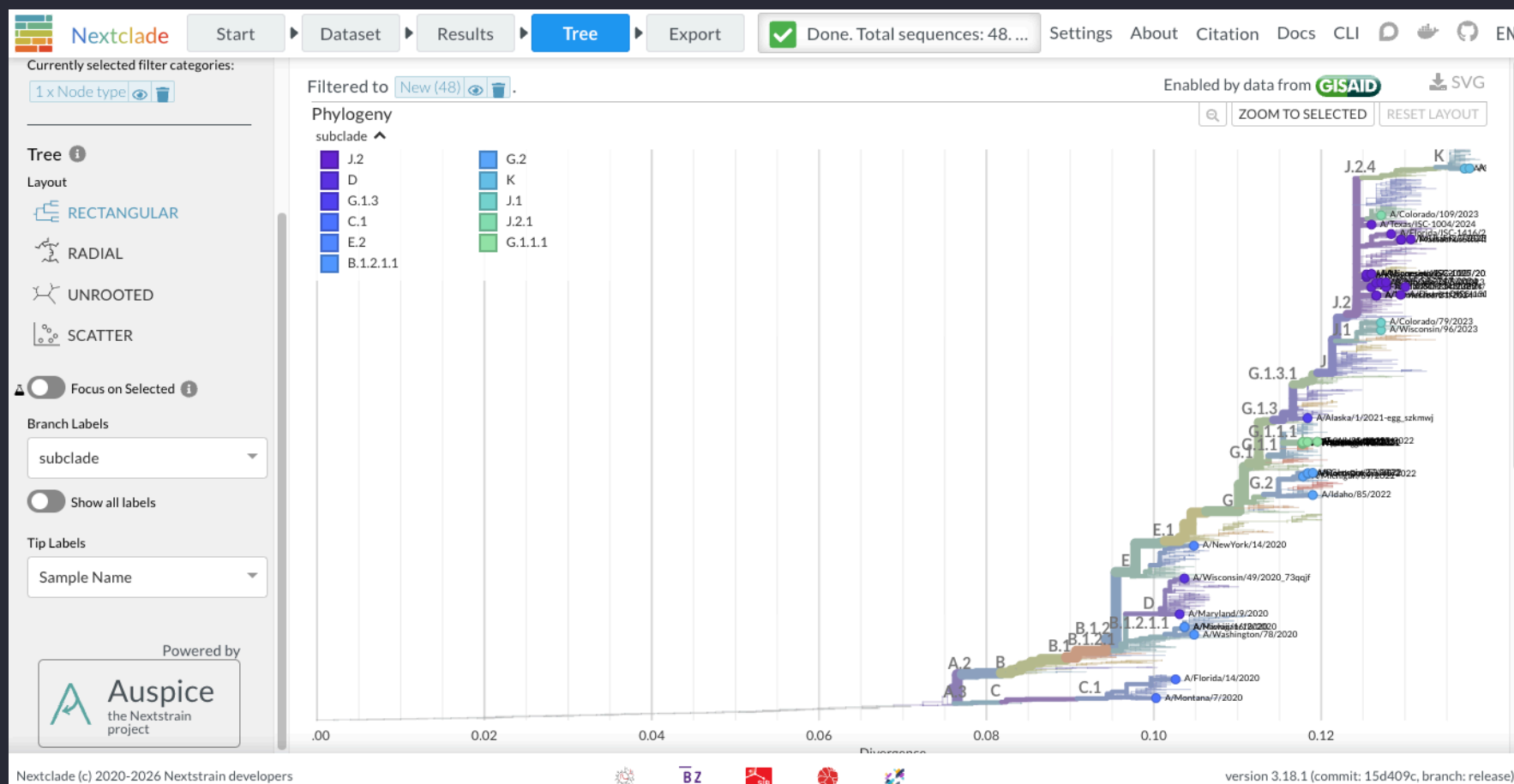
# The Concept of Clades

**Clades** are the fundamental grouping of phylogenetic trees. They are defined as a monophyletic group of taxa in a tree that includes a single common ancestor and all its descendants.

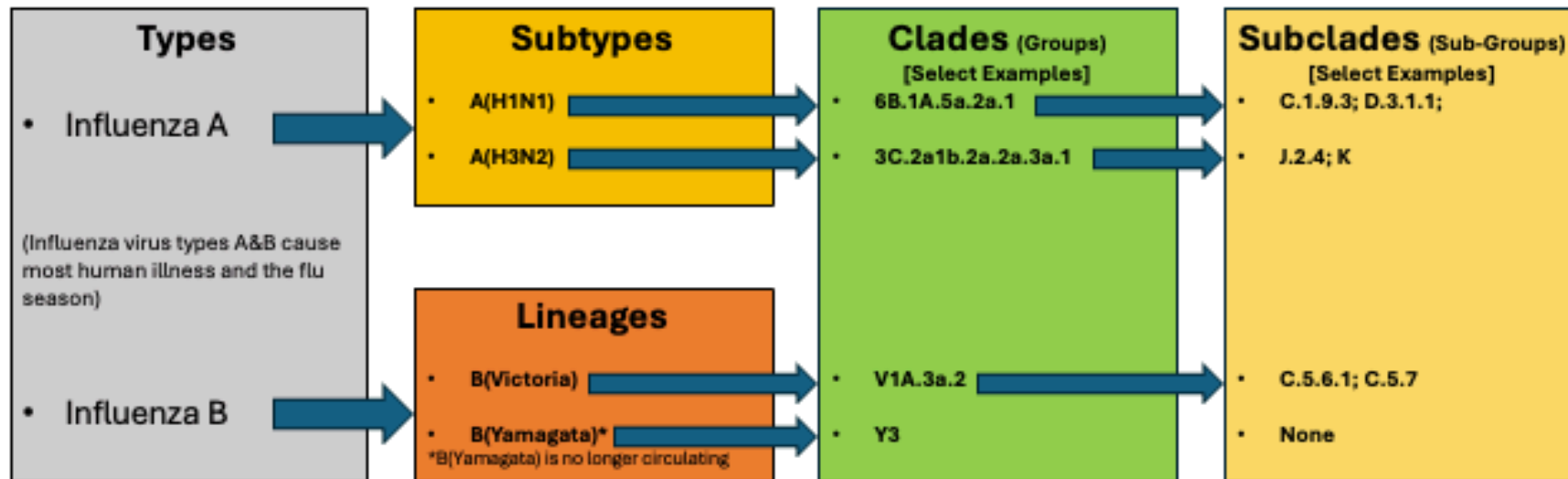


# Clades in Seasonal Influenza

The seasonal flu research community recently implemented a **nomenclature and clade proposal system** for tracking important emerging clades. These correspond to the “subclade” designations from **Nextclade** outputs.



# Clades in Seasonal Influenza (cont.)





# Clades in Seasonal Influenza (cont.)

Defining a subclade in Nextstrain (**augur clades command**):

| clade | gene | site | alt |
|-------|------|------|-----|
| A     | HA1  | 45   | N   |
| A     | HA1  | 48   | I   |
| A     | nuc  | 473  | T   |

| clade | gene  | site  | alt |
|-------|-------|-------|-----|
| K     | clade | J.2.4 |     |
| K     | HA1   | 2     | N   |
| K     | HA1   | 144   | N   |
| K     | HA1   | 158   | D   |
| K     | HA1   | 160   | K   |
| K     | HA1   | 173   | R   |

# Clades in Seasonal Influenza (cont.)

Subclade K example:

# Subclades vs. Lineages (Pango 🧐)

Influenza subclades are **not** equivalent to Pango (SARS-CoV-2) lineages.

Pango lineage designations only requirement is a group of viruses having a shared ancestry.

Lineage proposals are through public submission

This has led to the designation of 5.7K+ lineages. 🤯

# Subclade Proposal Process

Subclade proposal criteria:

1. Size: large groups should have a higher priority for designation.
2. Divergence: the more mutations have accumulated relative to the break point of the parent clade, the higher the priority of a novel clade
3. Specific mutations: Ideally, breakpoints sit on long branches with significant mutation. Such mutations will be better defined for well studied segments/genomes.

Each of these components feeds into the calculation of normalized branch scores across a phylogenetic tree.

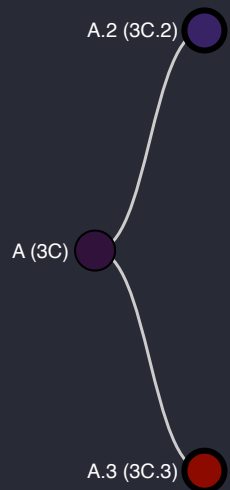
A threshold score of 1.0 is set.

# Subclade Naming

- The system of subclade naming uses a variation of the Pango nomenclature system.<sup>1</sup>
- Capital letters are used as aliases and numbers separated by periods for the retained part of the hierarchical name.
- Shortening (aliasing) of names after three hierarchical levels (i.e. D for C.1.1.1).
- However, a new alias may be given earlier in the case of a rapid expansion or special designation.

# Example of Implementation Using H3N2 HA

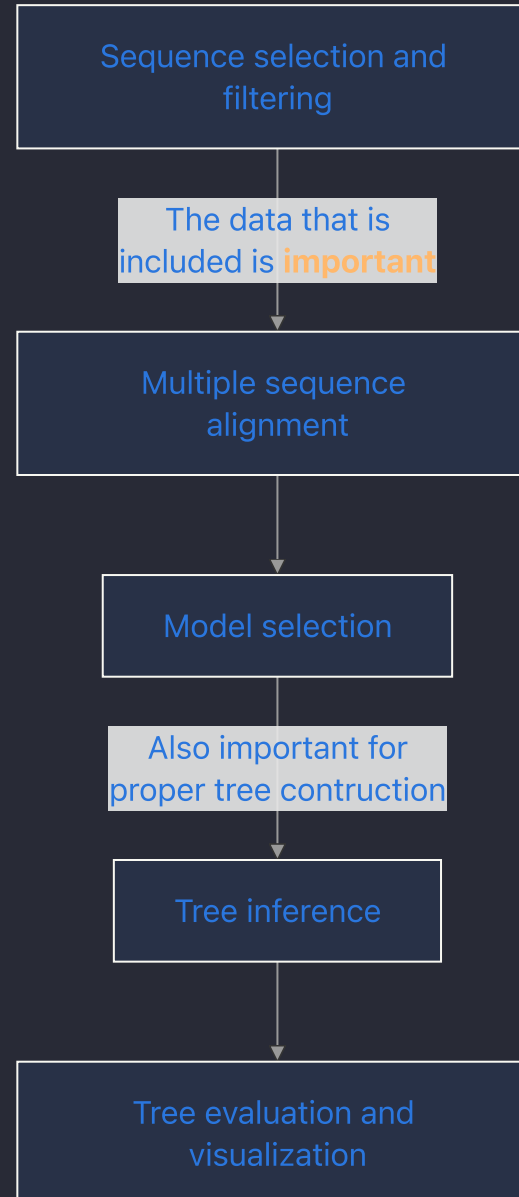
► Code



# Section 2: Tree Building Overview

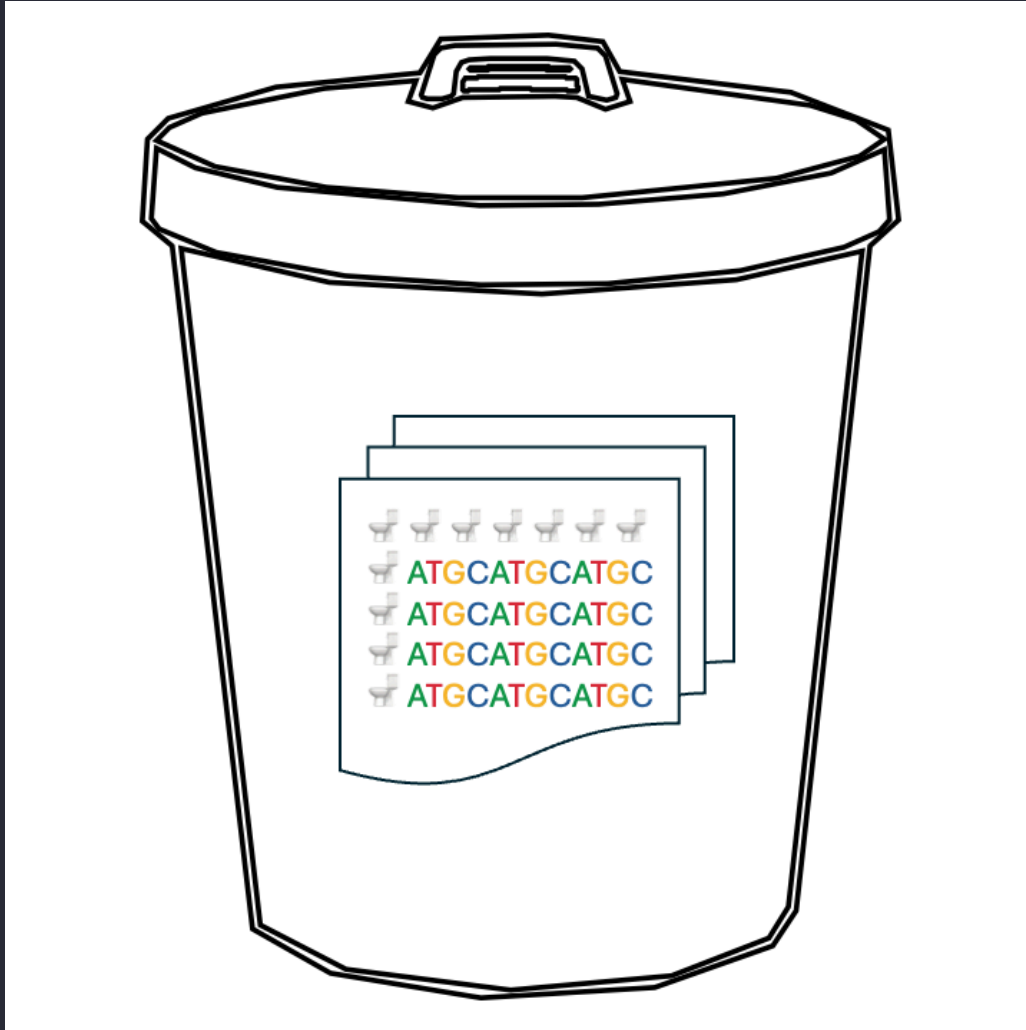
Quick break.

# Phylogenetic Tree Building Steps

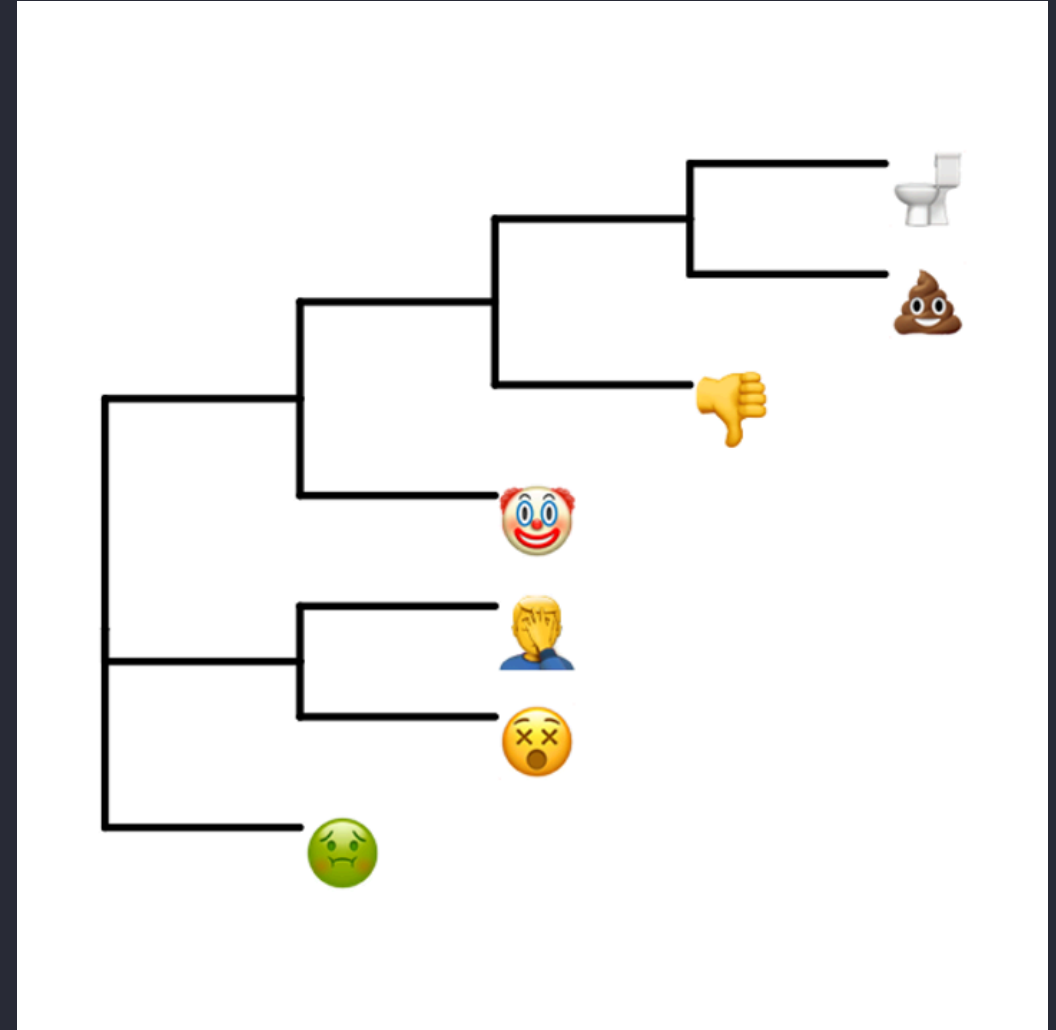




# Sequence Selection



Garbage in



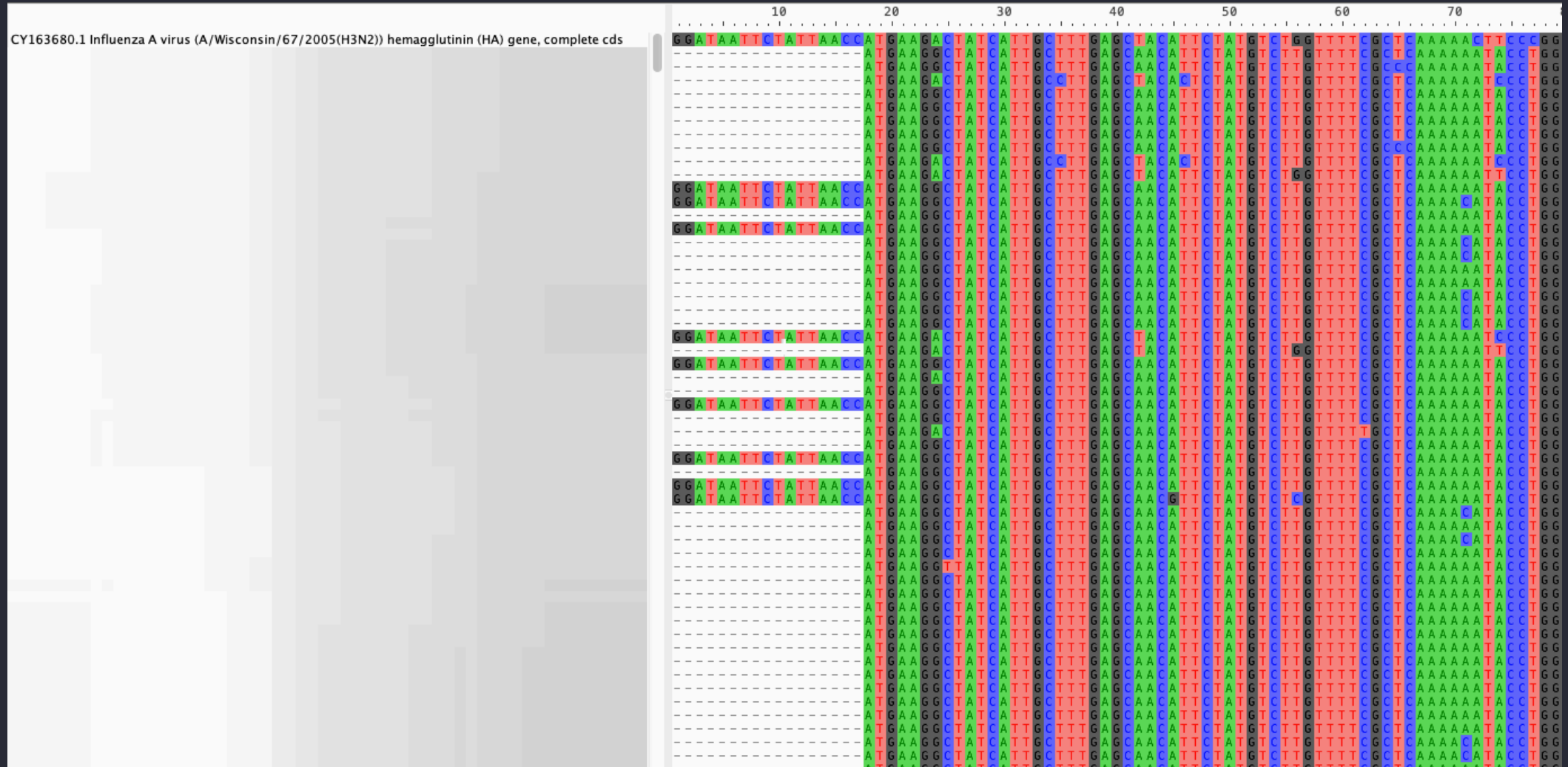
Garbage out



# Sequence Selection

When starting the process of building a tree:

# Multiple Sequence Alignment (MSA)



# MSA Algorithms

Examples of MSA software:

- CLUSTAL Omega
- MAFFT
- MUSCLE
- **nextclade** aligner (previously nextalign)

# Model Selection

Assuming nucleotide sequence data as input:

- Jukes-Cantor (simplest model)<sup>1</sup>
- HKY
- TN93
- GTR (Model implemented in Nextstrain flu pipelines)

# Tree Inference - Algorithms

## Distance Based:

- Neighbor joining
- UPGMA

## Character Based:

- Maximum Parsimony
- Maximum Likelihood (used in Nextstrain pipelines)
- Bayesian Inference

# Tree Inference - Software

- MEGA
- RAxML
- FastTree
- IQ-TREE (Nextstrain Default)
- cmaple (Standalone program and within IQ-TREE using `--pathogen` flag)



# Tree Evaluation and Visualization

- Nextstrain (auspice)
- [auspice.us](https://auspice.us) (browser based auspice)
- IcyTree (browser based)
- FigTree
- ITOL
- Dendroscope
- archaeopteryx
- Programmatic libraries:
  - ETE Toolkit
  - ggtree

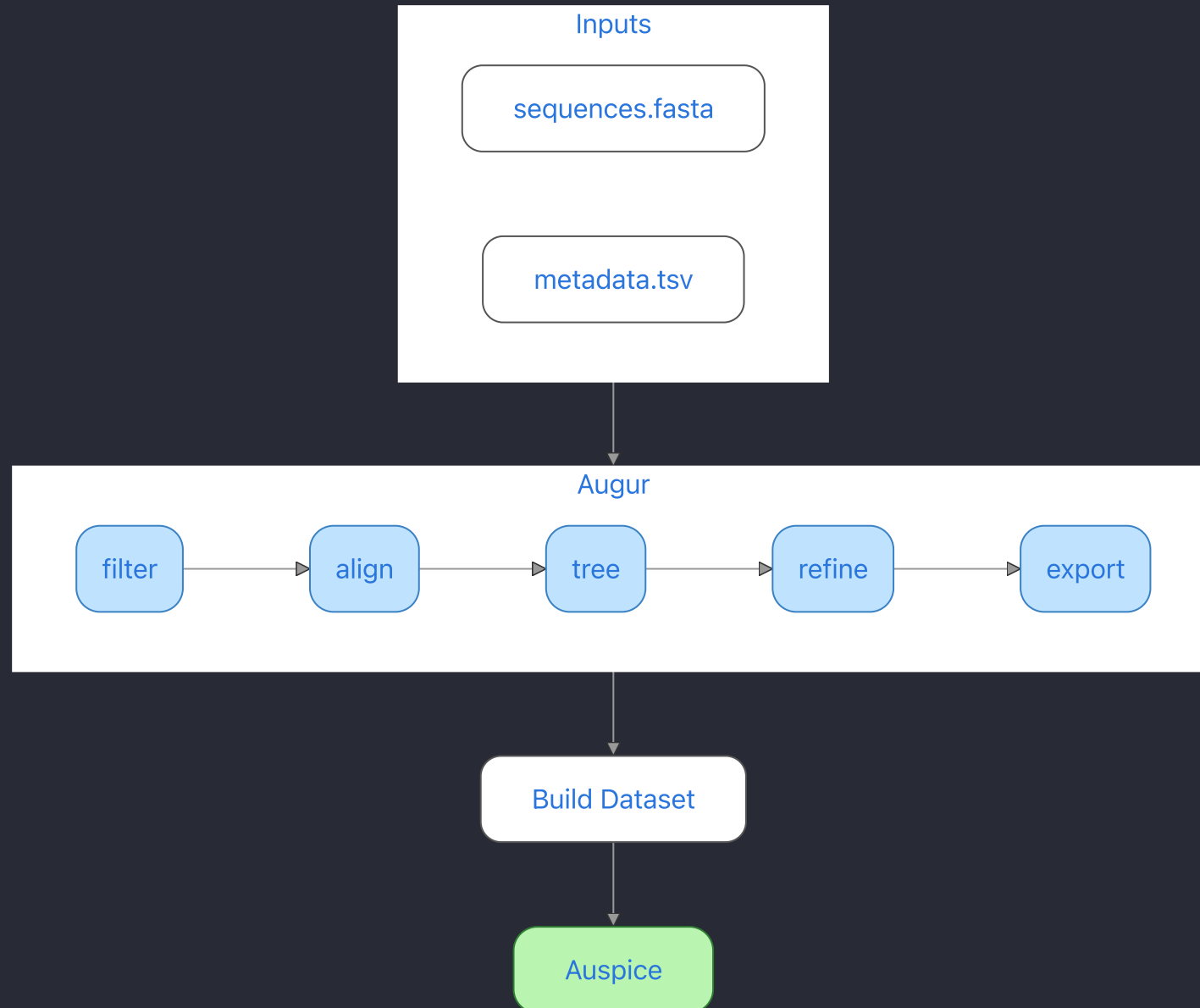
# Intro to Nextstrain

**Nextstrain** is an open source toolkit for analyzing and visualizing pathogen genomic data.

The two core parts of Nextstrain are:

- Augur (modular bioinformatics toolkit)
- Auspice (browser based visualization)

# Nextstrain process overview



# Section 3: Nextstrain Build Exercise

Click on the following [link](#) which will direct you to the exercise we will be going through together.

# Section 4: Overview of Nextstrain Features and Navigation

We'll go over features of your build and navigate it together.

# Section 5: Communicating Results Using Nextstrain Narratives

# Introduction to Nextstrain Narratives

Nextstrain narratives are markdown (.md) files that allow the communication of results interactively in a slide-like format.

Click on the link above for detailed documentation on narratives.

For an example of a narrative, take a look at the [Twenty years of West Nile virus](#) narrative.

# Components of a Narrative: Header

Open the example narrative file `/profiles/gisaids/custom.md` in VS Code.

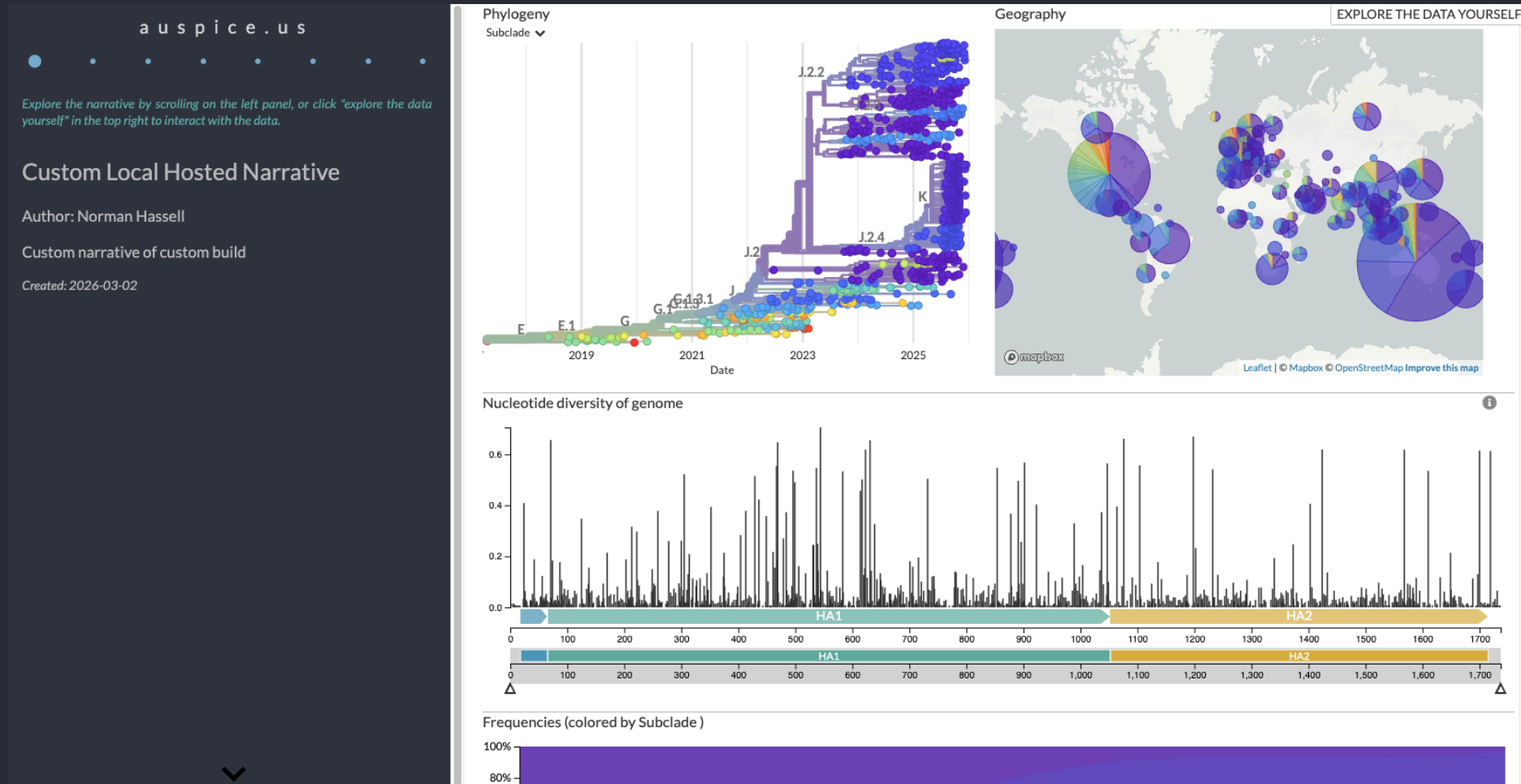
At the top of the file is the narrative header:

```
1 ---
2 title: Custom Local Hosted Narrative
3 authors: Norman Hassell
4 date: "2026-03-02"
5 dataset: "https://nextstrain.org/custom/h3n2/ha"
6 abstract: "
7 Custom narrative of custom build
8 "
9 ---
```

For local builds, you will have the prefix `https://nextstrain.org` followed by your build file name with underscores replaced by the `/` character (ie. `custom_h3n2_ha.json` dataset becomes `https://nextstrain.org/custom/h3n2/ha`)



# Components of a Narrative: Header (Result)

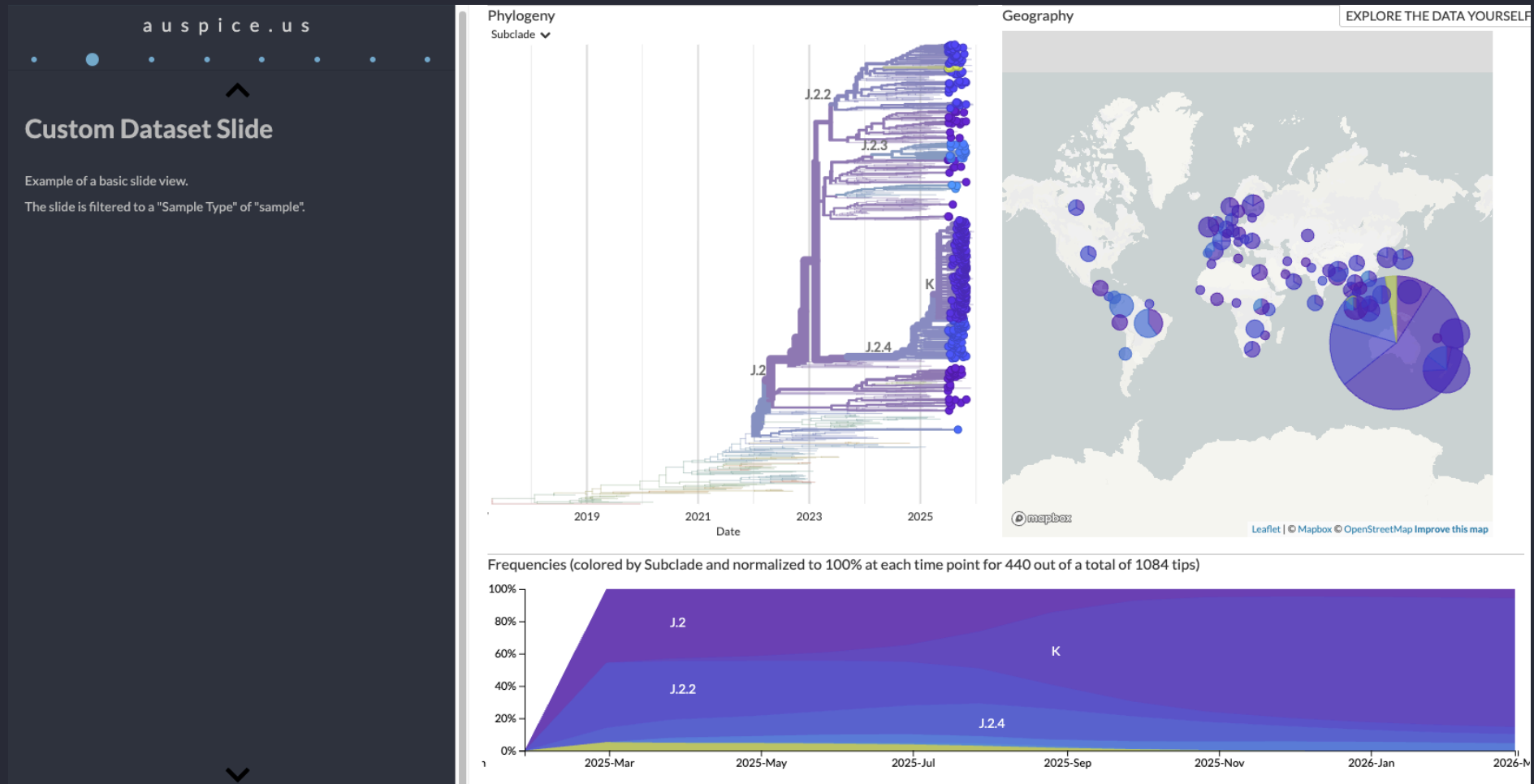


# Components of a Narrative: Basic Slides

Basic slides are set up by providing copied links as first level headers:

```
1 # [Custom Dataset Slide](https://nextstrain.org/custom/h3n2/ha?d=tree,map,f
2
3 Example of a basic slide view.
4
5 The slide is filtered to a "Sample Type" of "sample".
```

# Components of a Narrative: Basic Slides (Result)



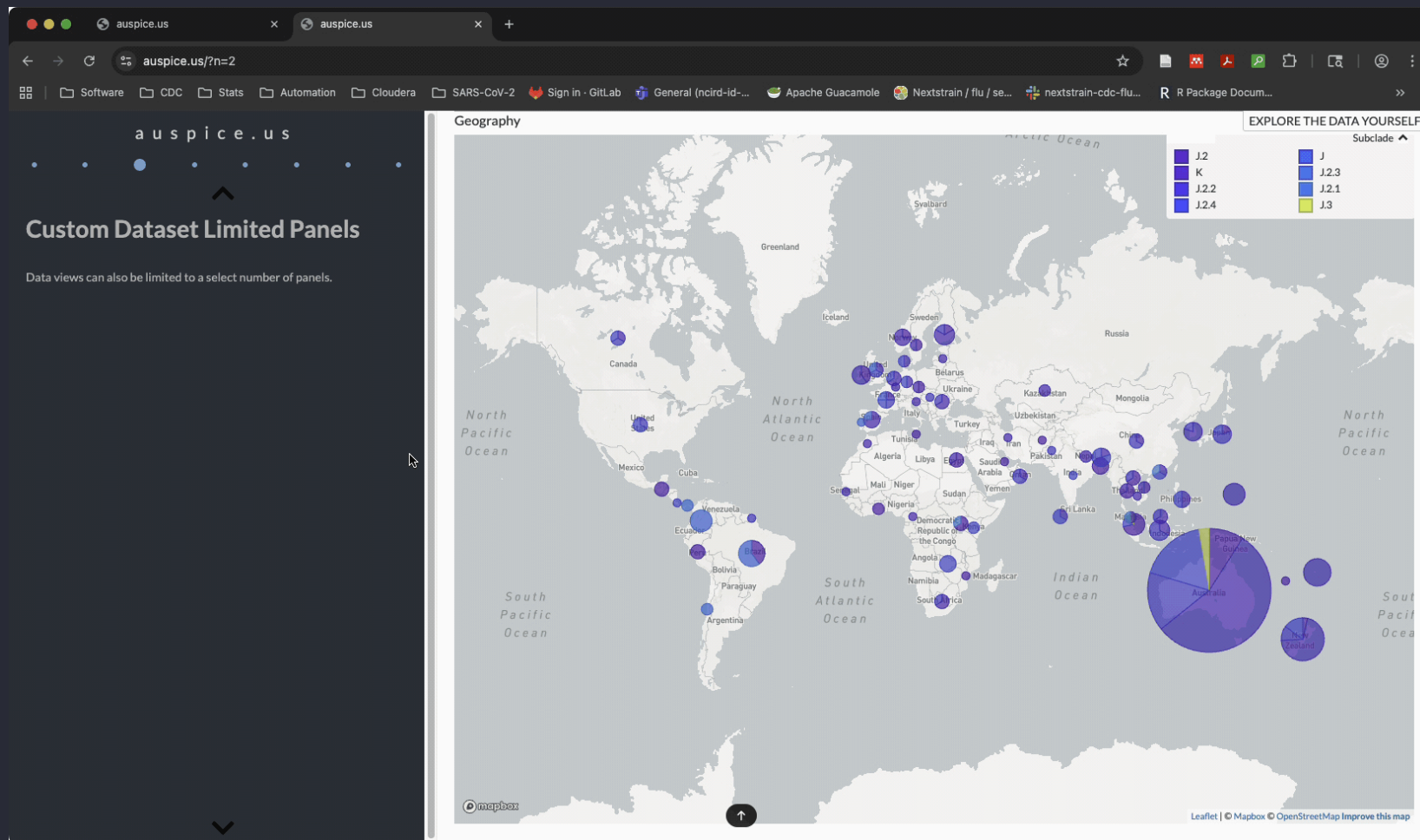
# Components of a Narrative: Animations

Timeline animations can be used in narratives by copying the link associated with the animation.

The link will briefly show after clicking the “Play ” button underneath the “Date Range” toolbar.

```
1 # [Custom Dataset Animations](https://nextstrain.org/custom/h3n2/ha?animate)
2
3 Animations can be captured.
```

# Components of a Narrative: Animations (Result)



# Components of a Narrative: Images

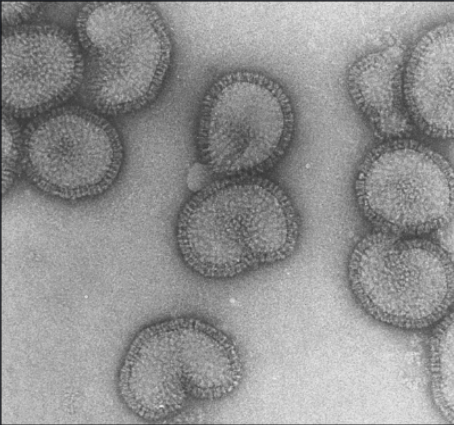
Images can be imbedded into the narrative sidebar or main display from direct links or **base64 conversion**.

```
1 # [Image Embedding](https://nextstrain.org/custom/h3n2/ha)
2
3 Images can be embedded from the web in the side panel.
4
5 

auspice.us

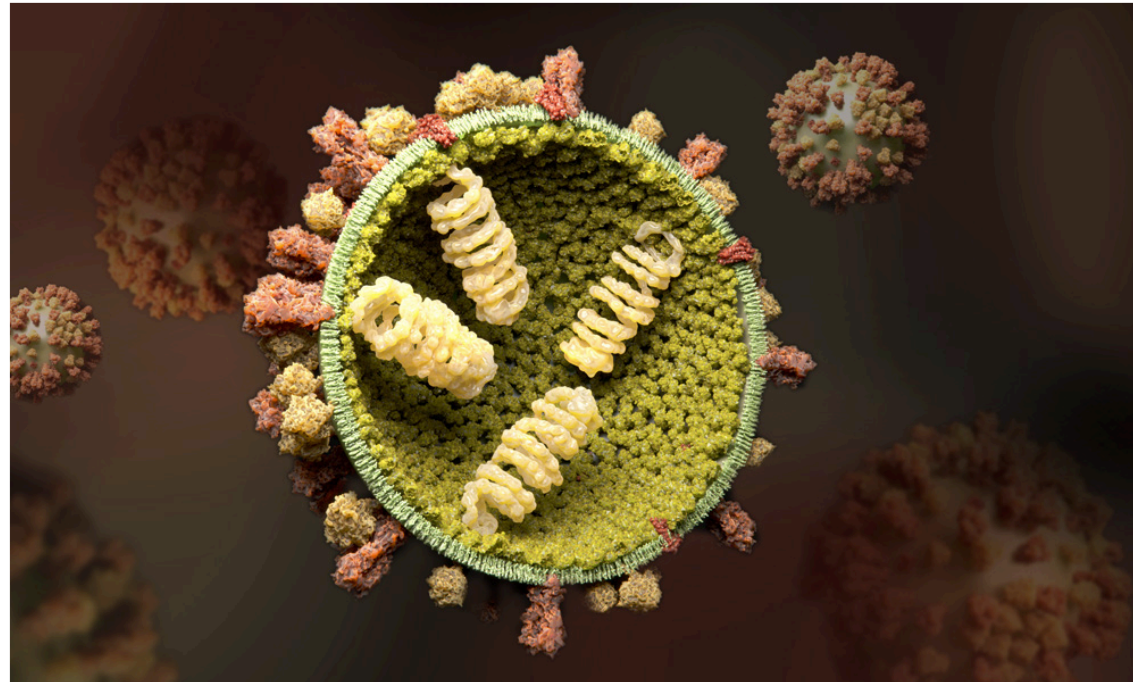
Image Embedding

Images can be embedded from the web in the side panel.



Or in the main body display via the `auspiceMainDisplayMarkdown` section.

For main body markdown display.



# Narratives Continued

Normal features of markdown such as:

- Tables
- Lists
- Text formatting
- HTML embedding

Can all be used in both the narrative side panel and main display.

Again, the main display requires the `auspiceMainDisplayMarkdown` block for use.



# Narrative Exercise

Before moving on to the exercise open a new terminal in the `seasonal-flu-demo` folder and enter into a new nextstrain shell session:

```
1 nextstrain shell .
```

Then, run another build using the following command:

```
1 nextstrain build . --configfile profiles/gisaid/custom_gisaid_global.yaml
```

Repeat this process for a third build.

Open a new terminal, enter a nextstrain shell, and execute the following:

```
1 nextstrain build . --configfile profiles/gisaid/custom_gisaid_norefs.yaml
```

# Narrative Exercise

Develop a narrative walking through your current build focusing on viruses circulating in Oceania/Australia (`custom_h3n2_ha`).

- What were the most prominent subclades for samples?
- What subclade is exhibiting the highest local branching index (LBI)?
- From the identified subclade with the highest LBI, what is the estimated emergence date and range?
- What regions is this subclade present in?
- What is the rate calculation for the tree overall vs. the rate for the subclade?
- Are there any reagent viruses in this subclade? Highlight them.
- Add another narrative slide of your choice, highlighting something you've found.
- Close out your narrative with a summary slide.

# Section 6: Build Comparisons

We'll take some time to compare the three different builds we've created.